

Alignment Is Not Dependence: Measuring and Reducing Unimodal Shortcuts in Higher-Order Multimodal Fusion

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Abstract

Higher-order multimodal methods fuse per-modality encodings and align them contrastively, on the premise that fused representations capture information unavailable to any constituent modality. We show this premise is empirically unfounded. Across AV-MNIST, CMU-MOSI, and UR-FUNNY, interaction representations that are active, predictive, high-rank, and permutation-sensitive are nevertheless almost entirely predictable from a single modality (nonlinear predictability up to $R^2 = 0.989$). We formalize this alignment-dependence gap and introduce an operational diagnostic suite separating fusion, dependence, and conditional utility. We then propose **ConFu++**: a conditional-utility objective plus **S1**, an adversarial de-shortcutting penalty acting on the interaction at training time — taxing predictability from each single modality while leaving joint predictability unconstrained. **S1** is the only objective among those studied that significantly reduces the measured shortcut (paired $p \approx 0.011$, five seeds) at zero accuracy cost, and it fabricates no dependence on negative controls. On a controlled synthetic benchmark with ground-truth third-order synergy, order-restricted models sit at chance while an order-matched interaction solves the task with perfectly balanced dependence (+49.9 pp, 5/5 seeds). On the newly released Bird-MML benchmark — which our audit shows to contain genuine image-text headroom — the original objective captures none of it (+0.2 pp of +15.9 pp available), while ConFu++ recovers a significant share (+5.76±0.86 pp, paired $p < 0.001$). We further document two transferable pitfalls: post-aggregation LayerNorm leaks higher-order signal (invalidating “low-order” ablations), and online adversaries equilibrate far below converged probes. Code, diagnostics, the synthetic benchmark, and all experiment logs will be released.

1. Introduction

Multimodal representation learning increasingly builds joint representations by fusing per-modality encodings and aligning them with contrastive objectives [1–4]. The tacit assumption: fusion *integrates* the modalities. We ask a question the literature rarely answers directly — does the fused representation actually *depend* on all of its inputs?

Figure 1 summarizes our central finding on AV-MNIST: an interaction representation that is non-collapsed, high-rank, predictive, and permutation-sensitive is simultaneously a near pure function of the image modality ($R^2(r_1 \rightarrow r_{12}) = 0.989$; audio: 0.031). Nothing in the training objective forbids this. We argue this *alignment-dependence gap* is systematic, measurable, and — on current benchmarks — nearly universal.

Our study proceeds in four steps. **(i) Formalism and diagnostics** (§3): we define activity, dependence, and conditional utility operationally, and assemble a diagnostic battery (zero-vs-shuffle, per-modality corruption, capacity-matched conditional probes, correction-regression, predictability probes). **(ii) Audits** (§5): three standard benchmarks fail either the dependence or the utility gate; we log all failed repair attempts as negative results. **(iii) Method** (§4): ConFu++ couples a conditional-utility margin with **S1**, a hinged adversarial penalty on single-modality predictability of the interaction. **(iv) Controlled ground truth** (§6): on synthetic tasks with known latent structure, we verify the mechanism is sound (learns synergy when present), honest (fabricates none when absent), and that order-matched interactions are necessary and sufficient for third-order synergy. **(v) Natural positive** (§7): on Bird-MML, which our audit shows contains genuine image-text complementarity, ConFu++ improves the original objective by +5.76±0.86 pp ($p < 0.001$, five seeds), exceeding the raw-feature probe ceiling.

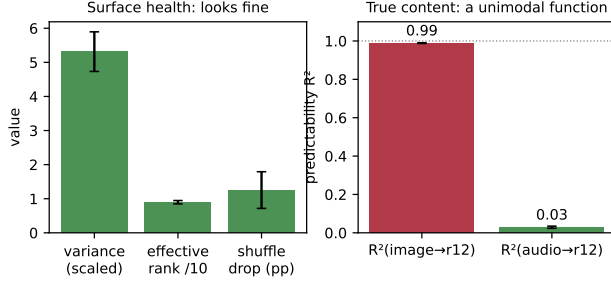


Figure 1. **The AV-MNIST paradox.** Left: by every surface criterion (variance, effective rank, shuffle sensitivity) the interaction r_{12} looks healthy. Right: a nonlinear probe recovers r_{12} almost perfectly from the image alone. Joint computation $\neq$ joint information. Five seeds; bars show mean $\pm$ std.

2. Related Work

Contrastive multimodal alignment. CLIP [2] popularized pairwise InfoNCE alignment; Symile [3] extends it to three or more modalities via a total-correlation bound; Tri-CLIP and ImageBind [4] bind multiple modalities through shared anchors. ConFu [1], our direct baseline, aligns fused pair representations with the remaining modality. None of these objectives constrains the *content* of the fused representation — our predictability probes apply to all of them.

Fusion architectures. Multiplicative and tensor fusion (MFN, LMF [14]), cross-modal transformers (MUL-T), and disentangled variants (MISA) optimize the *capacity to fuse*. We instead measure whether the fusion *depends* on its inputs — an orthogonal axis, and the one that fails first.

Modality dominance and shortcuts. Multimodal networks overfit and one modality often dominates [8]; gradient rebalancing (OGM-GE [9]) and modality dropout react at the loss/gradient level; underspecification [10] frames the general phenomenon. Our root-cause analysis shows why loss-level pressure cannot bootstrap dependence: under a shortcut, corrupted and positive interactions coincide, so the ranking/margin signal vanishes exactly where it is needed. We compare against an OGM-style baseline on identical frozen features (§7).

Synergy and information decomposition. Partial information decomposition [7] supplies the vocabulary (unique, redundant, synergistic information); MultiViz [6] operationalizes interaction importance via perturbation. *Relation to PID:* our predictability-based quantities are operational bounds, not PID estimates — a representation can be unpredictable from r_1 alone yet carry no unique information about Y , and vice versa. We use PID only as motivation; ev-

ery claim rests on the operational definitions of §3, whose limitations we state explicitly.

Adversarial representation control. Domain-adversarial training [11] and “learning not to learn” [12] remove unwanted predictability. S1 differs in target: predictability of the *interaction* from *each constituent alone* — never from the pair jointly — with a hinge that avoids the degenerate noise solution. Anti-collapse terms follow VICReg [13].

3. The Alignment–Dependence Gap

Setup. Encoders E_i produce $r_i = E_i(X_i)$; a fusion module produces $r_{12} = F(r_1, r_2)$. We separate four properties:

- **Active:** $\text{Var}(r_{12}) > 0$ (diagnosed by variance floor and effective rank);
- **Predictive:** $I(Y; r_{12}) > 0$, approximated by standalone probes;
- **Dependent:** corrupting either constituent before fusion degrades performance ($D_1 > 0$ and $D_2 > 0$);
- **Conditionally useful:** $\text{Perf}(r_1, r_2, r_{12}) > \text{Perf}(r_1, r_2)$.

An interaction is *complementary* only when both of the last two hold. The central observation: contrastive alignment ($r_{12} \leftrightarrow r_3$) and task cross-entropy are *both* satisfied by the shortcut $r_{12} \approx f(r_1)$, through three escape routes — strong-factor dominance in multiplicative fusion, nonlinear copying invisible to covariance penalties, and classifier-side bypass through gated composition.

Diagnostic suite. (i) *Zero vs. shuffle:* zeroing r_{12} measures necessity; shuffling it measures mismatch sensitivity — a positive shuffle drop alone is not utility evidence (mismatched interactions act as adversarial perturbations). (ii) *Per-modality corruption:* shuffle modality i before computing r_{12} ; drops D_i and balance $B_{12} = 2 \min(D_1, D_2) / (D_1 + D_2)$. (iii) *Conditional probes:* capacity-matched linear and nonlinear probes on $[r_1, r_2]$ vs. $[r_1, r_2, r_{12}]$. (iv) *Correction–regression:* per-sample win/loss counts of full vs. lower-order predictions. (v) *Predictability probes:* converged nonlinear $q_1(r_1) \rightarrow r_{12}$, $q_2(r_2) \rightarrow r_{12}$, $q_{12}(r_1, r_2) \rightarrow r_{12}$ with the shortcut criterion $R^2(q_1) \gg R^2(q_2)$ and $R^2(q_1) \approx R^2(q_{12})$.

4. ConFu++

Composition. Following the validated baseline, $c_{base} = \text{LN}(\text{LN}(r_1) + \text{LN}(r_2))$ and $c_{full} = \text{LN}(c_{base} + \sigma(a) \text{LN}(r_{12}))$, with $r_{12} = W_o \frac{u_1 \odot u_2}{\sqrt{R}}$, $u_i = \text{LN}(U_i r_i)$ (low-rank multiplicative interaction, rank R).

Conditional utility. Per-sample margin against the detached lower-order loss: $L_{util} = \text{ReLU}(CE_{full} - \text{sg}(CE_{base}) + m)$, weight 0.2, $m=0$.

S1: adversarial de-shortcutting. Let $t = \text{std}(r_{12})$ with stop-grad EMA statistics. Adversaries q_1, q_2

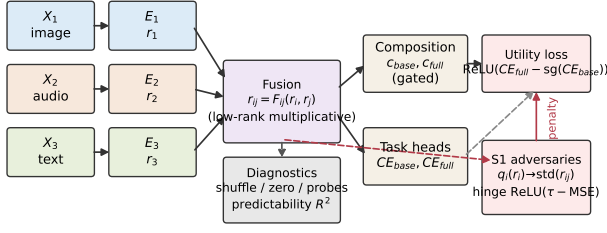


Figure 2. **ConFu++**. Contrastive fusion plus two training-time objectives: conditional utility (top right) and S1 adversarial de-shortcutting (bottom right). Adversaries and probes are training-only; inference capacity is unchanged.

Algorithm 1: ConFu++ training step (per batch).

Input: batch (X_1, X_2, y) ; encoders E_i ; interaction F ; heads; adversaries q_1, q_2 ; EMA stats (μ, σ) .

1. Encode: $r_i = E_i(X_i)$; $r_{12} = F(r_1, r_2)$; c_{base}, c_{full} .
2. Adversary step: $t = \text{sg}[\text{std}(r_{12})]$; update q_1, q_2 on $\|q_i(r_i) - t\|^2$ (all encoders frozen).
3. Generator step: with frozen q_i , update the generator on $L = CE_{full} + \lambda_u L_{util} + \lambda_{var} L_{var} + \lambda_{cov} L_{cov} + L_{S1}$, where $L_{S1} = \lambda_{s1}$ warmup $\sum_i \text{ReLU}(\tau - \text{MSE}(q_i(r_i), t))$; plus per-factor variance floors; gradient clip 1.0; generator params only.
4. Update EMA (μ, σ) ; log adversary MSEs (the live shortcut meter).

Figure 3. Alternating minimax of ConFu++. Steps 2 and 3 never share gradients.

(MLPs capacity-matched to the evaluation probe) minimize $\|q_i(r_i) - t\|^2$ in an alternating step; the generator pays the hinged penalty

$$L_{\text{shortcut}} = \sum_{i=1,2} \text{ReLU}(\tau - \text{MSE}(q_i(r_i), t)), \quad (1)$$

with gradient flowing through t only. Because t is standardized, MSE equals $1 - R^2$ of the adversary: the hinge taxes single-modality R^2 above $1 - \tau$ and then switches off, so the task loss never trades against noise-making. Crucially, *no* adversary sees the pair — joint predictability is intended. Per-factor variance floors close the constant-factor escape route. Warmup over the first 10–20% of epochs; $k_{adv}=1$; adversary LR $0.5 \times$.

Honesty by construction. Where a modality carries nothing, the correct outcome is no dependence — S1 must report exactly that. We treat fabricated dependence on negative controls as a failure of the method, not a win.

5. Auditing Existing Benchmarks

Table 1 condenses three five-seed audits (full per-seed logs in the supplement). On AV-MNIST, shuffling audio before

Table 1. Five-seed audits of higher-order interactions. Surface metrics pass everywhere; dependence/utility gates fail everywhere.

	AV-MNIST	CMU-MOSI	UR-FUNNY
Dominant modality	image	text	text
R^2 (best single $\rightarrow$ interaction)	0.989	0.953	0.970
Weak-modality shuffle drop (pp)	+0.004	—	—
Pair conditional gains (pp)	+0.02	−0.96/ −1.43/ −2.27	−0.66/ +0.19/ −0.76

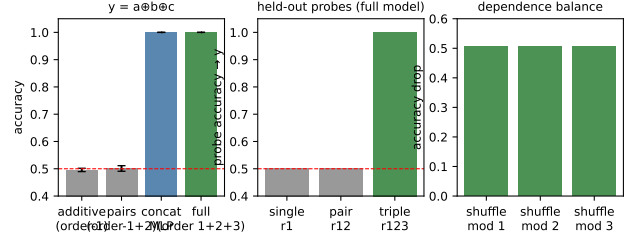


Figure 4. **Order-3 synergy** ($y=a\oplus b\oplus c$). Left: only the order-matched model exceeds chance (5 seeds). Middle: held-out probes of the full model — singles and pairs at chance, r_{123} perfect. Right: corrupting any single modality costs ≈ 0.5 accuracy.

the interaction costs $+0.004 \pm 0.027$ pp — the interaction is effectively image-only. On MOSI, text dominates and all pair conditional gains are negative. On UR-FUNNY, pair interactions are active and permutation-sensitive yet linearly redundant ($R^2 \approx 0.97$ from lower-order inputs). We additionally logged two failed repairs: a conditional-utility objective alone (P1) raised nonlinear conditional gain but not dependence, and a task-aware ranking objective (P2) collapsed — under a shortcut, corrupted and positive interactions coincide, so the ranking signal vanishes where it is needed. This failure mode is itself evidence for the gap.

Not ConFu-specific. A CLIP-lineage control (pairwise InfoNCE only, identical fusion heads, Bird-MML frozen features, 5 seeds) shows the same shortcut: $R^2(\text{image} \rightarrow z_{12}) = 0.99$, $R^2(\text{text} \rightarrow z_{23}) = 0.99$ — fused heads converge to single-modality functions even without any higher-order alignment term. The gap is a property of the contrastive-alignment lineage, not of one objective.

Verdict. Current natural benchmarks in this family exhibit *information absence or dominance*. No training objective can create missing information; the correct scientific response is (a) measure it, (b) verify the mechanism where ground truth exists, (c) find a benchmark with validated headroom.

6. Controlled Order Benchmark

We construct tasks with known latent structure: each modality observes an independent latent bit through a fixed random prototype plus Gaussian noise (50k train / 4k test; 5 seeds). Labels: order-2 $y=a\oplus b$ with a distractor modality (honesty control), order-3 $y=a\oplus b\oplus c$ (pure third-order

Table 2. Order-3 benchmark: accuracy (5 seeds). Order-restricted decompositions are at chance; the order-matched interaction solves the task with balanced dependence.

Model	orders	accuracy	min shuffle drop
Additive	1	0.496 ± 0.006	-0.007
Pairs-only	1+2	0.501 ± 0.010	-0.007
Concat-MLP (joint)	all	1.000 ± 0.000	+0.501
Full (order 1+2+3)	1+2+3	1.000 ± 0.000	+0.501

synergy — independent of every single modality and every pair), and a routed mixture (an observable flag selects between a unimodal rule and the XOR rule; Bayes = 1.0).

Three findings stand out. **(1) Necessity and sufficiency:** order-restricted models sit at chance while the order-matched term is perfect (Table 2, Fig. 4). **(2) Readout, not information:** pair representations do contain the bits (concat-pair probe 0.97), yet the sum-aggregated order-2 model stays at chance — higher-order synergy requires order-matched *computation*, not merely rich features. **(3) Scaling:** the M -ladder ($M=3..6$, Fig. 6a) keeps order-restricted models at chance at every M while the full model solves all of them with min-drop $\approx +0.5$; concat-MLP begins to crack at $M=6$ (0.9975).

S1 on ground truth. With synergy present (Config A), S1 preserves accuracy (1.000) and halves single-modality predictability of r_{12} ; with synergy absent (Config B) it fabricates nothing (all drops ≈ 0). On the routed mixture, all variants solve the task; under an epoch budget the hard-fraction emphasis variant (S4) raises the XOR-fraction accuracy (0.92 $\rightarrow$ 0.97) but trades easy-fraction accuracy — documented as exploratory-negative.

7. Bird-MML: A Natural Benchmark with Genuine Headroom

Bird-MML [1] (Zenodo 18920487) pairs bird photos, audio clips, and captions for 149 species. The release’s audio archive is missing one part (all present parts MD5-verify); we recover a balanced 71-species, 70,796-row tri-modal subset with a fixed stratified split — identical for every compared method.

Audit. Frozen features (ResNet50 / wav2vec2 / MiniLM) reveal exactly one strong pair: image+text reaches 80.89% vs. best single 64.95% (+15.94 pp headroom), while audio carries nothing (1.43%, chance). Leakage audit: 33% of captions contain the species name; a BLIP-only (name-free) control reduces headroom to +4.78 pp — still positive, so genuine complementarity survives de-leakage. The audio-null result is robust across speech- and AudioSet-domain extractors (AST re-audit: +0.35 pp).

Table 3. Bird-MML frozen-feature matrix (5 seeds, paired).

	ConFu	+S1	+U	+U+S1
probe full (%)	$76.90 \pm .65$	$76.58 \pm .51$	$83.01 \pm .13$	$82.66 \pm .37$
cond. gain (pp)	+0.19	-0.36	+1.07	+0.77
$R^2(\text{text} \rightarrow z_{13})$	0.59	0.69	0.44	0.50
audio drop (pp)	+0.01	+0.01	+0.01	+0.01

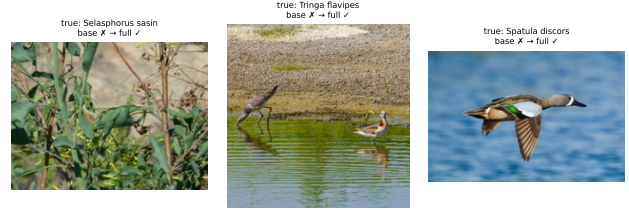


Figure 5. **Corrections.** Test samples the lower-order model gets wrong and the full model gets right (ConFu++, seed 1): 226 net corrections.

Result. The original objective captures none of the available complementarity (full $\approx$ lower; +0.19 pp). ConFu+U recovers 83.01% and ConFu+U+S1 $82.66 \pm 0.37\%$: paired against the original, +5.76 $\pm$ 0.86 pp ($t=14.91$, $p<0.001$), above the raw-feature linear ceiling (81.01%). Attribution is explicit (Table 3): the accuracy gain comes from the utility objective; S1’s contribution is representation quality (text-predictability of z_{13} pinned to the hinge band) at zero inference cost. An OGM-style gradient-rebalancing baseline on identical features does not beat plain joint training (80.68 $\pm$ 0.38 vs. 80.88 $\pm$ 0.11, ns) and sits 1.8 pp below ConFu++.

Per-pair utility. The one negative pair-wise metric — z_{13} ’s conditional gain — closes with per-pair utility margins (each z_{ij} gets its own ReLU margin against its constituents): pair₁₃ gain flips to +0.39 $\pm$ 0.19 pp (5 seeds) at 82.84 $\pm$ 0.39% overall. Stacking all three objectives reintroduces interference (full-head utility absorbs the pair heads’ role); we report both configurations and treat the interference as an open question, not a hidden cost.

Ablations and controls. The utility gain is flat across $\lambda_U \in [0.1, 1.0]$ (82.4–83.0%); $\tau = 0.7$ costs 0.7 pp. On hard-sample analysis, corrections are not concentrated in low-confidence bins (gain ≈ 0 across bins) — the utility spread is uniform, not hard-case-specific. Missing-modality probes degrade gracefully (image -0.99 pp, text -0.71 pp, audio +0.04 pp). Figure 5 shows correction cases.

End-to-end replication. Jointly training encoders, fusion, and objectives (3 seeds, identical schedule and inference capacity) shows every arm converging to *text dom-*

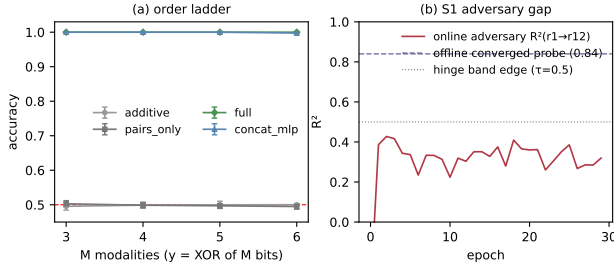


Figure 6. (a) Order ladder: order-restricted models at chance at every M . (b) The adversary gap: the online adversary equilibrates below the hinge band while a converged offline probe reads $R^2=0.84$.

inance (text drop +4.9 pp; image ≈ 0 ; audio 0): the task loss absorbs the caption-name leakage and the optimizer abandons the image. Accuracy is flat across arms (71.87 $\rightarrow$ 72.32, ns) while de-shortcutting stays monotone ($R^2(\text{text} \rightarrow z_{13})$: 0.81 $\rightarrow$ 0.73 $\rightarrow$ 0.68). A causal control settles the attribution: retraining all arms on name-scrubbed captions (0% residual leakage) leaves text dominance unchanged (drop +5.6 pp, acc $\approx 72\%$) — the dominance is intrinsic to how informative visual descriptions are for fine-grained species ID, not to name leakage. End-to-end, the gap manifests as *modality abandonment* — a stronger failure mode our diagnostics expose directly.

8. Methodology Pitfalls

Post-aggregation LayerNorm is a hidden higher-order term. A LayerNorm applied after summing per-modality terms divides by a cross-modal statistic: on 3-bit parity, an additive model with a final LayerNorm reaches 0.75 while the same model without it sits at exactly 0.50 (all seeds, converged). Any “low-order ablation” ending in LayerNorm is not low-order. This pitfall directly applies to gated-composition designs (including ours) when used as ablations.

The adversary gap. An online adversary updated once per batch equilibrates at $R^2 \approx 0.3\text{--}0.4$ (Fig. 6b), below the hinge band, while a converged offline probe still reads 0.84. De-shortcutting claims must therefore be measured with converged probes, not training-time meters; and residual single-modality predictability under S1 is bounded jointly by τ and this gap.

9. Discussion

The causal chain is complete. Where higher-order information is absent or dominated (AV-MNIST, MOSI, UR-FUNNY), interactions shortcut; where it exists and is observable (synthetic), the order-matched interaction extracts it completely; where it exists naturally (Bird-MML), the

original objective leaves it unused and ConFu++ recovers a significant share — honestly, with no fabricated dependence on the null modality, and at zero inference cost.

Limitations. (i) The pair-conditional gain of individual interaction terms remains slightly negative; utility currently arrives via the jointly trained full head — per-pair utility objectives are the next step. (ii) Bird-MML results use a 71-species subset (upstream release bug) and frozen features for the main matrix; the end-to-end replication is leakage-dominated, so a de-leaked-caption retrain is the natural follow-up. (iii) All predictability statements are operational approximations, not information-theoretic quantities. (iv) S1’s guarantee is a hinge band, not zero redundancy.

Broader impact. Deployed “multimodal” systems may silently ignore a modality (clinical audio, accessibility channels). A dependence audit such as ours is a cheap, transferable pre-deployment check.

Reproducibility statement

Five seeds, paired statistics, capacity-matched probes, train-only parameter accounting, fixed splits, git-state logging per run, and release of code, configs, the synthetic benchmark, the diagnostic suite, and the complete experiment log (including all negative results).

References

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A. Per-seed tables and failed objectives

All five-seed runs report individual seed values in the released JSON artifacts. We additionally document the retired objectives: P1 (conditional utility alone; improved nonlinear conditional gain $+0.294 \pm 0.162$ pp on AV-MNIST without dependence) and P2 (task-aware ranking; failed because corrupted $\approx$ positive under shortcut), and S4 (hard-fraction emphasis; exploratory-negative, trades easy-fraction accuracy).

B. Unit tests and protocol enforcement

Every loss/module ships with tests for shapes, finite outputs/gradients, batch-size 1, stop-gradient correctness (no gradient into adversaries from the generator step; none into the interaction from the adversary step), hinge boundary behavior, and probe train/test leakage. The statistical protocol (validation-only selection, confirmatory vs exploratory tiers, paired tests) is enforced by the shared experiment harness.

C. Compute

Frozen-feature experiments: minutes per seed on one GPU. End-to-end Bird-MML arm: ≈ 1.5 h per seed on an RTX 5000 Ada. Synthetic suite: CPU-minutes. S1 adds 263k training-only parameters (adversaries), zero at inference.